Project Roadmap: one strong project per role

These are projects to build next. They are not on the resumes yet. Add each one to the matching resume only after it runs and is pushed to GitHub with a README.

Java Full Stack / Backend: Hospital Appointment and Slot Booking System

Stack: Spring Boot, Spring Data JPA, Hibernate, Spring Security + JWT, MySQL, React, Swagger, JUnit, Docker Compose
What it solves and shows: Clinics lose time to double-booked slots. Role-based login (patient, doctor, admin), slot locking so two people cannot book the same slot, email reminders, paginated search.

QA / Testing: E-commerce Test Automation Framework

Stack: Selenium WebDriver, Java, TestNG, Page Object Model, Rest Assured, Maven, Extent Reports, GitHub Actions
What it solves and shows: A hybrid framework against a public demo store: 40+ manual test cases with RTM, 20+ automated UI tests, API tests, SQL data checks, and a bug report file with screenshots.

Python Developer: Expense Tracker REST API

Stack: Django, Django REST Framework, PostgreSQL, JWT, pandas, pytest, Docker
What it solves and shows: Multi-user expense API with categories, monthly budgets, CSV import cleaned with pandas, summary endpoints and 80%+ test coverage. Add a small Flask version of one service to show both frameworks.

AI / ML Engineer: Customer Churn Prediction with Explainability

Stack: Python, pandas, scikit-learn, XGBoost, SHAP, FastAPI, Streamlit, Docker
What it solves and shows: Telecom churn dataset: EDA, feature engineering, compare 3 models with cross-validation, explain predictions with SHAP, serve the best model through an API and a dashboard.

Generative AI Developer: Document Q&A; Assistant (RAG)

Stack: Python, LangChain, ChromaDB or FAISS, embeddings, an LLM API, FastAPI, React
What it solves and shows: Upload PDFs, chunk and embed them, answer questions with cited page numbers, and show an evaluation sheet of 25 questions with correct/incorrect results.

Frontend / MERN: Real-Time Kanban Task Board

Stack: React, TypeScript, Redux Toolkit, Tailwind CSS, Node.js, Express.js, MongoDB, Socket.IO, JWT
What it solves and shows: Drag-and-drop boards, live updates between two browsers, auth, filters, dark mode, Lighthouse score above 90. Covers the missing completed MERN project.

Database / SQL Developer: Retail Sales Data Warehouse

Stack: MySQL or PostgreSQL, star schema, stored procedures, triggers, window functions, indexing, Power BI
What it solves and shows: Design a star schema from a public sales dataset, write 25 analytical queries, show before/after execution plans for 3 slow queries, and build a Power BI report on top.

DevOps Engineer: CI/CD Pipeline for the Job Portal

Stack: Docker, Podman, Docker Compose, GitHub Actions, Jenkins, AWS EC2, Nginx, Kubernetes (Minikube), Bash
What it solves and shows: Dockerfiles for API, front end and DB; a pipeline that builds, tests, pushes images and deploys to EC2; Kubernetes manifests; the same images run under Podman; a README with an architecture diagram.

For every repository

- README with problem statement, screenshots, setup steps and an architecture diagram.
- A Dockerfile and docker-compose.yml so anyone can run it with one command.
- A live demo link or a 2-minute screen recording.
- At least a few tests and a clean commit history.